QUENTIN MORAINE

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PERSONAL STATEMENT

Engineering student completing a double degree between Icam, France, and the MSc Robotics at Cranfield University, with hands-on experience in machine vision, embedded systems and thermal-hydraulic engineering. Led a four-person team developing a vision-based sorting machine, then delivered the design basis of a cooling system during a 12-week internship at a district-heating operator. International exposure through an Erasmus semester in Malta, an internship in La Réunion and postgraduate study in the UK. Methodical, records every assumption so results can be checked, and adapts as readily to a construction crew as to an engineering office.

KEY ACHIEVEMENTS

- Achieved a mean average precision of 0.979 in screw head-type recognition by training a YOLO11n model, as project lead of a four-person team (Icam, 2026).
- Avoided about 37 kW of unnecessary chiller capacity by demonstrating the cooling plant had to be sized on summer rather than winter load (Réseaux de Chaleur Urbains d'Alsace, 2026).
- Secured a place on the double-degree programme leading to the MSc Robotics at Cranfield University (Icam, 2026).
- Achieved a TOEIC English score of 950 out of 990 (ETS, 2026).

EDUCATION

MSc Robotics: Cranfield University, UK **September 2026 - September 2027**

- **Modules:** Fundamentals of Robotics, Artificial Intelligence for Robotics, Robotics Control, Programming Methods for Robotics, Psychology, Ethics and Human Factors, Robotic Perception and Systems Engineering, Human-Robot Interaction, Autonomy in Robotic Systems
- **Group Project in Digital Robotics:** team-based research project from March 2027, applying the taught modules to a real-life problem, followed by a seven-month industrial placement
- **ABB RobotStudio Project:** built an IRB 120 drawing cell matching the lab setup and generated a path of about 220 targets from a CAD model of a logo, then checked reach, robot configurations and approach moves in simulation before handing in the station and its RAPID program

Diplôme d'Ingénieur: Icam Strasbourg-Europe, France **September 2022 - September 2027**

- **Award:** Master's-level General Engineering degree awarded as ECAM Arts et Métiers after the school joined the Icam group in 2022
- **Modules:** Control Systems, Embedded Systems, Electronics, Instrumentation, Finite Element Analysis, Fluid Mechanics, Thermal Engineering, Vibrations
- **Technical Project:** led the development of an automated screw-sorting machine within a group of four, combining YOLO11n head-type classification, OpenCV length and diameter measurement and Raspberry Pi 5 actuator control
- **Erasmus Exchange:** one semester at the Malta College of Arts, Science and Technology in 2023

CAREER HISTORY

Réseaux de Chaleur Urbains d'Alsace, Strasbourg: Engineering Intern **June 2026 - September 2026**

Regional operator of 30 district-heating networks with 140 km of pipework and a turnover of 70 million euros. Internship within the 15-person design office.

- Calculated the design basis of a closed-loop cooling system to replace an open-loop circuit prohibited by French environmental regulations, at a plant serving 40,000 dwelling-equivalents.
- Compiled a traceable heat balance from poorly documented equipment, setting the design point at 85 kW winter load, 17.7 m³/h total flow and free cooling below 14 °C.
- Demonstrated the chiller had to be sized on summer rather than winter load, avoiding about 37 kW of unnecessary installed capacity.
- Detected six inconsistencies across source documents and corrected four before release, matching the systems integrator's figure to within 0.3%.
- Produced the hydraulic schematics in BricsCAD and handed over ten deliverables, including four tender documents issued to contractors.

Grands Travaux de l'Océan Indien, La Réunion: Site Intern **June 2025 - August 2025**

Civil engineering company of the Colas Group. Six-week internship on a refrigerated logistics warehouse built for the client Logistisud in Saint-Pierre.

- Installed utility networks and dug trenches on a worksite where around ten contractors worked simultaneously.
- Shadowed the site manager for a full week, tracking progress, coordinating contractors and resolving on-site issues.
- Accompanied a land surveyor for a full day of surveying work.
- Applied site safety procedures daily in a demanding outdoor environment far from an academic setting.

KWS, Tours: Agricultural Worker, then Seasonal Team Leader **July 2020 - August 2020**

International seed company. Summer job during secondary school.

- Removed maize tassels by hand across more than 10 seed production plots over two months.
- Supervised a team of 8–10 seasonal workers, allocating tasks, monitoring pace and reporting daily to management.

SKILLS, INTERESTS & EXTRACURRICULAR ACTIVITIES

- **Languages:** native French and fluent English at C1 level with a TOEIC score of 950, plus German and Spanish at A2 and basic Italian.
- **IT Skills:** Python with OpenCV, Ultralytics YOLO, NumPy, Pandas and Flask. C++ with object-oriented programming and the Standard Template Library. TypeScript, Git and Linux.
- **Robotics and Embedded Systems:** ABB RobotStudio offline programming in RAPID, covering work objects, tool definition, AutoPath and a Smart Component gripper for pick and place. ROS 2 at an introductory level. Raspberry Pi, ESP32 with PlatformIO and Arduino for sensor and actuator interfacing.
- **Design Software:** SolidWorks, Autodesk Inventor, Autodesk Fusion and BricsCAD for mechanical design. KiCad and EasyEDA for electronics. Basic MATLAB.
- **Software Development:** developed Finanza, a desktop finance application of about 4,800 lines of TypeScript on Next.js, Prisma, SQLite and Electron, which imports statements from 2 banks. Code at github.com/quentin-moraine.
- **Embedded Development:** started C++ firmware for an ESP32 air-quality monitor reporting carbon dioxide, temperature and humidity to Apple HomeKit.
- **Student Union:** served students, staff and visiting lecturers daily at the cafeteria of a campus of 600, handling sales and restocking from February 2025 to March 2026.
- **Campus Events:** staffed logistics, set-up and front-of-house service at events for around 100 attendees.
- **Volunteering:** raised funds annually for a guide-dog charity throughout secondary school.
- **Individual Interests:** gym, volleyball with a university club, cycling, photography and video editing with DaVinci Resolve and Lightroom.
- **Driving Licence:** full EU licence, category B.